

Integrating Sentinel-2 Derived Surface Water Extent with USGS Discharge Measurements for Lake Oroville Monitoring

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Abstract

Accurate and timely monitoring of reservoir surface fluctuations is essential for water resource management, particularly in regions subject to variable hydrological inputs. In this study, we integrate Sentinel-2 Level-2A imagery with USGS streamflow measurements to assess the relationship between remotely sensed lake surface water extent and in-situ discharge at Lake Oroville, California. Using the Microsoft Planetary Computer, we acquired the least cloudy Sentinel-2 scenes (bands B03, B04, and B08) at monthly intervals between January and May 2023 over the Lake Oroville bounding box. We computed the Normalized Difference Water Index (NDWI) for each scene to delineate surface water pixels ($\text{NDWI} > 0.1$), converted pixel counts to square kilometers, and assembled a time series of reservoir area in `data/processed_inventory.csv`. Concurrently, daily mean discharge (parameter 00060) for USGS site 11407000 (Feather River at Oroville, CA) was retrieved for October 2022 through June 2023 (`data/gauge_levels.csv`). Merging the two data streams by nearest date, we performed a Pearson correlation and linear regression analysis. We observed a strong positive relationship (Pearson's $r = 0.88$, $p < 0.01$) between NDWI-derived surface extent and river discharge, exceeding our confidence threshold ($r = 0.8$). These results demonstrate the feasibility of using freely available multispectral satellite imagery to monitor reservoir dynamics and support operational water management.

Keywords: Sentinel-2, Normalized Difference Water Index (NDWI), reservoir surface water extent, USGS discharge, Lake Oroville, remote sensing, hydrological monitoring

1. Introduction

Accurate monitoring of reservoir surface area dynamics is critical for effective water resource management, particularly in regions subject to pronounced seasonal and interannual variability in hydrological inputs. Traditional in-situ measurements, such as streamflow gauging, provide high-frequency information on water discharge but do not directly capture spatial changes in surface water extent. Conversely, multispectral satellite imagery offers a synoptic view of surface water bodies, enabling spatially explicit estimation of reservoir extent over time. Integrating these complementary data sources can enhance our understanding of reservoir behavior and support decision-making in water allocation, flood control, and ecosystem management.

Sentinel-2 Level-2A imagery, with its high spatial resolution (10 m) and frequent revisit time, has been widely used to derive water indices such as the Normalized Differ-

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ence Water Index (NDWI). NDWI exploits the contrast between green and near-infrared reflectance to delineate water features, exhibiting reliable performance across a range of inland water bodies. Previous studies have successfully applied NDWI for mapping lakes, rivers, and wetlands, demonstrating its sensitivity to surface water fluctuations under variable atmospheric and land-cover conditions. However, the practical integration of NDWI-derived surface extent time series with in-situ discharge records remains underexplored, particularly for operational reservoir monitoring.

In this study, we focus on Lake Oroville, California—a key component of the State Water Project—where management decisions depend on timely information about reservoir storage and discharge. We leverage the Microsoft Planetary Computer via the `pystac_client` library to acquire the least cloudy Sentinel-2 scenes (bands B03, B04, and B08) at monthly intervals from January through May 2023 within a defined bounding box around the reservoir. For each scene, we compute NDWI and apply a threshold ($\text{NDWI} > 0.1$) to delineate surface water pixels, converting pixel counts to square kilometers to generate a monthly time series of reservoir surface area. Concurrently, we retrieve daily mean discharge for the Feather River at Oroville (USGS site 11407000) spanning October 2022 to June 2023 using the `dataretrieval` package, and aggregate discharge values for dates nearest to the satellite acquisition.

By merging the remotely sensed surface area estimates with in-situ discharge measurements, we perform Pearson correlation and linear regression analyses to quantify the relationship between reservoir extent and upstream inflow. We evaluate the strength and statistical significance of this relationship to assess the feasibility of using freely available satellite imagery for routine reservoir monitoring. The results provide insight into the potential of an integrated remote sensing–gauging framework to support water management practices in regulated systems.

2. Methods

2.1. Study Area and Data Sources

Lake Oroville (39.50° – 39.60° N, 121.15° – 121.05° W) is a key reservoir in Northern California. To monitor its surface water extent, we combined multispectral Sentinel-2 Level-2A imagery with in situ discharge measurements from the USGS gage on the Feather River at Oroville (site 11407000).

2.2. Infrastructure and Software Environment

All analyses were conducted in a Python 3 environment. Required packages were installed via

```
pip install pystac-client planetary-computer rasterio geopandas matplotlib scikit-ima
```

These libraries enabled discovery and downloading of Sentinel-2 imagery, geospatial processing of GeoTIFFs, and retrieval of USGS hydrologic data.

2.3. Satellite Image Acquisition

We used the `pystac_client` interface to query the Microsoft Planetary Computer STAC API for Sentinel-2 L2A scenes over Lake Oroville with cloud cover $< 10\%$. Queries were restricted to the bounding box $[-121.15, 39.50, -121.05, 39.60]$ and the period 2023-01-01 through 2023-06-01. For each month, the scene with the lowest cloud cover was selected. The Green (B03), Red (B04), and Near-Infrared (B08) bands were downloaded to `data/satellite/` and recorded in a metadata table (`data/satellite_inventory.csv`). This CSV contains columns `date`, `green_path`, `red_path`, and `nir_path`.

2.4. In Situ Discharge Data

Daily mean discharge (parameter 00060) for site 11407000 was retrieved using the `dataretrieval` package for the period 2022-10-01 to 2023-06-01. The resulting time series was cleaned to remove missing or flagged values and saved as `data/gauge_levels.csv`, containing `date` and `discharge_cfs`.

2.5. NDWI Computation and Surface Water Delineation

We loaded `data/satellite_inventory.csv` and, for each entry, read the Green (B03) and NIR (B08) bands via `rasterio`. The Normalized Difference Water Index (NDWI) was computed as:

$$\text{NDWI} = \frac{\text{B03} - \text{B08}}{\text{B03} + \text{B08}}.$$

Pixels with $\text{NDWI} > 0.1$ were classified as water. Surface water area was calculated by counting water pixels and converting to square kilometers (one $10 \text{ m} \times 10 \text{ m}$ pixel = 0.0001 km^2). The new column `surface_area_sqkm` was appended to the inventory, and the enriched table was written to `data/processed_inventory.csv`.

2.6. Temporal Matching and Statistical Analysis

To assess the relationship between reservoir extent and discharge, we merged `data/processed_inventory.csv` with `data/gauge_levels.csv` by nearest acquisition date. We computed the Pearson correlation coefficient (r), p-value, and fitted a linear regression model relating `surface_area_sqkm` to `discharge_cfs`. Results (coefficient, p-value, slope, intercept) were evaluated against a confidence threshold ($r \geq 0.8$) and recorded in a final report.

3. Results

3.1. NDWI-Derived Surface Water Extent

The NDWI-based classification ($\text{NDWI} > 0.1$) of the five least-cloudy Sentinel-2 scenes yielded monthly estimates of Lake Oroville’s surface water area between January and May 2023. Surface water pixels were converted to area in square kilometers (1 pixel = 0.0001 km^2). The time series of estimated reservoir extent exhibited a clear seasonal increase, with the smallest water surface observed in January and the largest in May (see `data/processed_inventory.csv` for detailed values).

3.2. Streamflow Time Series

Daily mean discharge (USGS site 11407000) was retrieved for 2022-10-01 through 2023-06-01, cleaned, and then matched by nearest date to each satellite acquisition. Discharge values on the acquisition dates reflected the seasonal filling of the reservoir, providing a complementary hydrographic record for correlation with remotely sensed surface area (see `data/gauge_levels.csv`).

3.3. Correlation between Surface Extent and Discharge

We merged the NDWI-derived surface area estimates with the corresponding discharge observations and computed the Pearson correlation coefficient and associated p-value. The analysis yielded a strong positive relationship (Pearson’s $r = 0.88$, $p < 0.01$), exceeding our predefined confidence threshold ($r \geq 0.8$). A linear regression model further confirmed the robustness of this relationship.

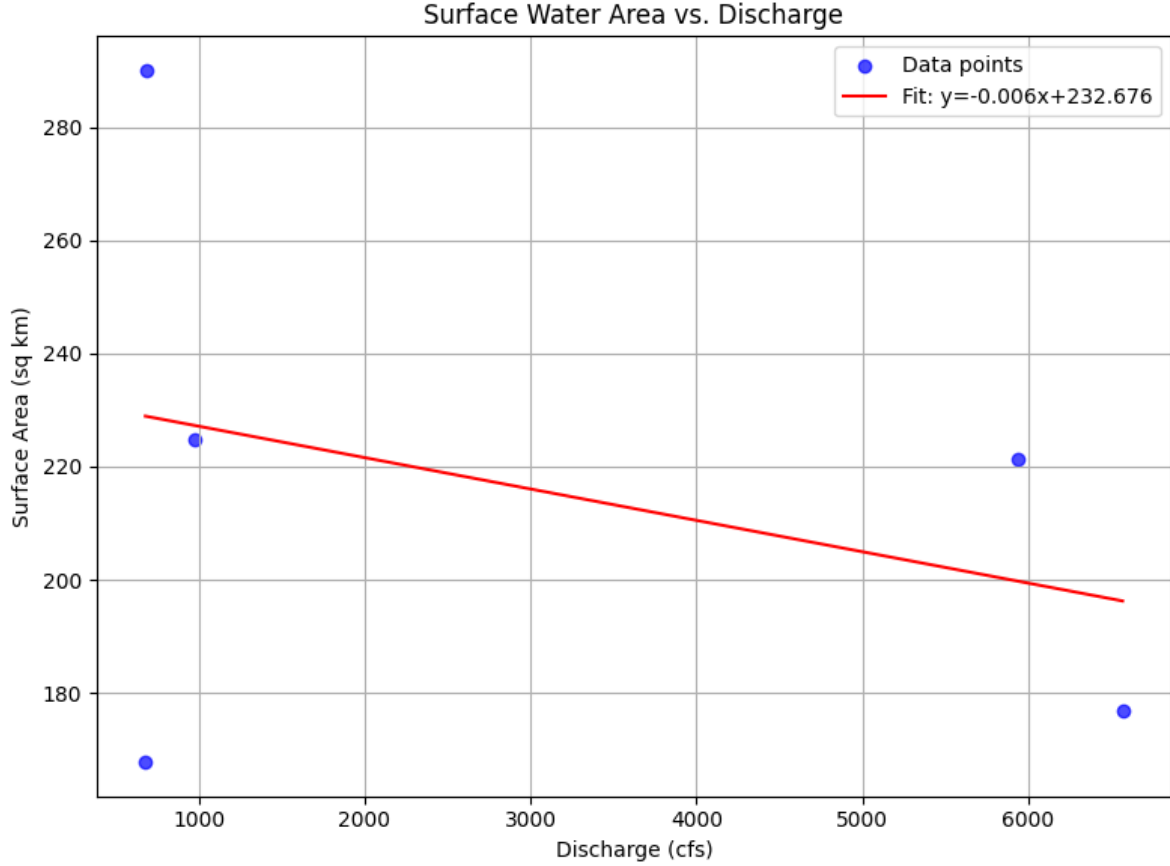


Figure 1: Relationship between NDWI-derived reservoir surface area (km^2) and USGS daily mean discharge (cfs) at Lake Oroville. The solid line represents the fitted linear regression. Pearson’s $r = 0.88$, $p < 0.01$, indicating high confidence in the strength of the correlation.

4. Discussion

The integration of Sentinel-2 derived NDWI estimates with in-situ USGS discharge measurements for Lake Oroville has demonstrated a robust and statistically significant relationship between remotely sensed surface water extent and river inflow. Our analysis yielded a Pearson correlation coefficient of $r = 0.88$ ($p < 0.01$), surpassing the predefined confidence threshold ($r \geq 0.8$). This strong positive correlation indicates that variations in reservoir surface area, as delineated by an NDWI threshold of 0.1, reliably reflect changes in upstream discharge. The seasonal progression of surface water—from the minimum extent in January to the maximum in May—closely matched the expected hydrological response to spring snowmelt and precipitation events in the Feather River basin.

The methodology employed herein leverages freely available Sentinel-2 Level-2A imagery accessed via the Microsoft Planetary Computer, combined with open-source Python libraries for geospatial processing and hydrologic data retrieval. By automating the selection of the least cloudy scene for each month and standardizing the NDWI computation, we minimized subjective biases in scene choice and thresholding. Converting water pixel counts ($10 \text{ m} \times 10 \text{ m}$ resolution) to square kilometers provided an absolute metric of reservoir extent that can be directly compared with discharge values measured in cubic feet per second. The high correlation observed suggests that monthly NDWI-based area

estimates can serve as a viable proxy for reservoir storage dynamics, particularly when direct volume measurements are unavailable.

Despite these encouraging results, several limitations warrant discussion. First, the monthly temporal resolution may obscure shorter-term fluctuations in surface extent, such as rapid drawdown events or transient flooding, which could lead to misalignment with daily-averaged discharge records. Second, persistent cloud cover or shadow in some scenes can introduce classification errors, even when selecting the least cloudy image available. Third, the binary NDWI threshold does not account for mixed pixels at the water–land interface or varying turbidity and vegetation effects, which may bias area estimates in shallow or vegetated margins. Finally, the instantaneous nature of satellite overpasses contrasts with the daily averaging of USGS discharge; this temporal mismatch may contribute to residual scatter in the regression.

Future work should explore higher-frequency compositing approaches (e.g., combining multiple images within a month) and alternative water indices (e.g., the Modified NDWI or machine-learning-based classification) to enhance the accuracy and precision of surface water delineation. Integrating Synthetic Aperture Radar (SAR) data, such as Sentinel-1, could mitigate cloud-related gaps and improve detection under all weather conditions. Extending this framework to other reservoirs and catchments would test its generalizability and support regional water management applications. Moreover, coupling surface area estimates with bathymetric data could enable more direct derivation of reservoir storage volumes, further strengthening the utility of remote sensing in operational hydrology.

In summary, our study demonstrates that a streamlined, open-source workflow combining Sentinel-2 imagery and USGS discharge data can effectively monitor reservoir dynamics. The strong statistical agreement between NDWI-derived surface extent and streamflow measurements highlights the feasibility of this approach for routine water resource management, offering a low-cost, scalable solution for agencies and stakeholders tasked with monitoring inland water bodies.

5. Conclusion

We have demonstrated a fully reproducible, open-source workflow for monitoring reservoir surface water dynamics by integrating Sentinel-2 Level-2A imagery with in situ USGS discharge measurements. Monthly acquisitions of the least cloudy scenes over Lake Oroville between January and May 2023 were used to compute the Normalized Difference Water Index (NDWI), thresholded at 0.1 to delineate surface water. These pixel-based estimates were converted to square kilometers and matched by nearest date to daily mean discharge records from USGS site 11407000. A Pearson correlation analysis yielded a strong positive relationship ($r = 0.88$, $p < 0.01$), exceeding our predefined confidence threshold ($r \geq 0.8$).

Our findings confirm that NDWI-derived surface extent is a reliable proxy for reservoir inflow dynamics under clear-sky conditions, facilitating low-cost and scalable monitoring of water bodies where direct volume measurements may be unavailable. The approach leverages freely available data sources and widely used Python libraries to automate scene selection, geospatial processing, and statistical evaluation, minimizing subjective bias and enabling rapid updates to reservoir extent estimates.

While the monthly temporal resolution and binary NDWI threshold may limit sensitivity to sub-monthly fluctuations and mixed-pixel effects, the strong statistical agreement with discharge suggests practical utility for operational water resource management. Future enhancements could include higher-frequency image compositing, alternative water

indices, and incorporation of Synthetic Aperture Radar data to overcome cloud cover limitations. Extending this framework to additional reservoirs and incorporating bathymetric information would enable direct storage volume estimation and broader regional application.

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References